

An Analysis on Cervical Cancer Risks and Early Detection

Chan Ga Wai
Faculty of Computing and Informatics
Multimedia University
Cyberjaya, Malaysia
1221305898@student.mmu.edu.my

Abstract—Cervical cancer screening is essential to early disease detection but manual Pap smear can be time-consuming and affected by observer differences, creating a need for assistive methods that can classify images and provide visual evidence for review. This paper presents a deep learning framework for cervical cytology image classification and pre-cancerous cell localisation using the RIVA dataset. A baseline convolutional neural network, ResNet34 and EfficientNet-B0 were trained and compared for binary classification of pre-cancerous and non-lesion images. The same models were also extended to six-class, seven-class and eight-class multi-class classification. For localisation, U-Net and Stacked Hourglass were developed to identify the location of pre-cancerous cervical cells using heatmap and keypoint outputs. The best binary classifier was also used as a point-level classifier gate in the final integrated prediction pipeline. The results in this research shows that deep learning models can support cervical cytology image classification and localisation, while visual outputs can help explain model behaviour.

Keywords—cervical cytology, deep learning, classification, CNN, ResNet, EfficientNet, U-Net, Stacked Hourglass, localisation, medical image analysis

I. INTRODUCTION

Cervical cancer is ranked as the fourth most common cancers that impacts the livelihood of women globally with an approximated 660000 new cases along with 350000 deaths in 2022 [1], [2]. Based on World Health Organisation, a global elimination strategy has also been introduced and it highlights the purpose of extensive screening [1]. However, while early detection through cytology-based screening like Pap smears remains crucial and necessary, conventional manual examination is similarly tedious, laborious and inclined to human error [3], [4].

With modern approaches in artificial intelligence, deep learning approaches present favourable options for further improvements of automated analysis. Deep learning models can learn visual patterns directly from cervical cytology images instead of depending only on handcrafted features. This is useful because cervical cytology images contain subtle morphological information, and different abnormal categories may share similar visual characteristics [5], [6].

The purpose of this research is to develop and evaluate a deep learning method for cervical cancer risk detection from cervical cytology images using the RIVA dataset. This study aims to train and compare several classification models, detect cervical cancer cells using cell localisation and examine their performance using conventional metrics such as precision, recall, F1-score, confusion matrices and mean Average Precision. Additionally, visualisation is used to decipher

model predictions and provide actionable insights for clinical screening.

Combining classification and localisation allows the method to not only distinguish whether cervical cytology images contain risky cells, but also visually highlight the abnormal cell locations. This helps to enhance interpretability because the model output does not only show a class prediction, but also shows the suspected regions in the image. The output of this research is only for research evaluation and serves no clinical diagnostic claims or validations for the methods used. The research is guided by the research objectives below:

Objective 1 : To develop, train, and compare multiple deep learning-based image classification models, including convolutional neural network (CNN), ResNet and EfficientNet for binary classification to detect pre-cancerous and non-lesion cervical cells from cervical cytology images while using performance metrics to evaluate the performances.

Objective 2 : To develop, train, and compare the same deep learning-based image classification models used previously for binary classification for multi-class classification for six, seven and eight labels which includes different combinations of pre-cancerous cells as well as at risk cells such as ENDO and INFL cells.

Objective 3 : To develop detection model for cell localisation based on pre-cancerous categories of cervical cells using best performing deep learning image-based classification as the main model with U-Net for heatmap regression and Stacked Hourglass for keypoint localisation of pre-cancerous cervical cells.

Objective 4 : To provide explanations and interpretations of the results obtained from the applied deep learning using visualisation and analytical techniques.

II. RELATED WORKS

A. RIVA Dataset and Bethesda System

The RIVA dataset is a high-definition accumulation of conventional Pap smear cytology images created to support artificial intelligence training [9]. It composes of 959 mini-patches of 1024 x 1024 pixels images that were extracted from whole-slide scans of 115 patient samples digitised at 40 times magnification. The dataset contains 26158 total annotations and 15949 unique cells. These diagnostic labels and keypoint annotations make RIVA suitable for both image classification and cell localisation.

RIVA includes five pre-cancerous classes and three non-lesion classes, consistent with the Bethesda classification. The

pre-cancerous labels include ASCUS, ASCH, LSIL, HSIL and SCC. The non-lesion labels include NILM, ENDO and INFL [10], [11]. The Bethesda System categorises cervical cytology findings to stratify the risk of pre-cancerous or cancerous changes and guide clinical management.

Keypoint annotation is also important in this research because it marks important points of interest inside the image. In cervical cytology, this supports models in learning where key cell locations occur. Instead of only assigning a class label to an image, keypoint annotations provide spatial information that can be converted into heatmap targets. This allows the localisation model to learn where the abnormal cell is likely to appear.

B. Deep Learning Models for Classification

CNNs has been the foundation of deep learning in image analysis because they can hierarchically learn spatial features, such as nuclear morphology and chromatin texture, which are vital in cytology [5], [6]. In cervical cancer research, CNNs have been employed for binary classification of Pap smear cells and they function as a dependable baseline for evaluating more advanced architectures [12], [13].

ResNet was introduced to facilitate training of very deep neural networks by incorporating skip connections that eases vanishing gradient as well as degradation problems [14]. In medical image analysis, including cervical cytology, different ResNet models have shown good performance for differentiating subtle morphologic disparities among lesion types under varied staining conditions [15], [16]. Therefore, ResNet is an important candidate for the comparative evaluation of deep learning models for pre-cancerous and non-lesion cell detection.

EfficientNet provides a compound scaling method that proportionally scales network depth, width and resolution. This can help to provide strong accuracy with less parameters and compute demands compared to other conventional deep networks [17]. The parameter efficiency of EfficientNet makes it suitable for medical imaging datasets that are limited and imbalanced [18], [19]. This supports its inclusion as one of the main models for classification experiments in this research.

Multiclass classification refers to assigning an image into one of multiple possible categories instead of only separating images into normal and abnormal groups. In cervical cytology, this task is difficult because different abnormal cell types may have similar visual features such as enlarged nuclei, darker chromatin, irregular nuclear shape, and changes in the nucleus-to-cytoplasm ratio. A model may perform well on majority classes while failing to recognise rare but clinically important abnormal cells. This issue is relevant to RIVA because its eight Bethesda classes are unevenly distributed with minority classes having fewer samples than NILM and INFL.

C. Cell Localisation

U-Net architecture remains one of the most widely adopted architectures for medical image segmentation because of its encoder-decoder structure and skip connections that preserve spatial details across feature scales [20]. In this

research, U-Net is adapted for heatmap regression so that the output can represent likely pre-cancerous cell regions.

Hourglass networks were originally made for keypoint prediction and are able to perform multi-scale keypoint detection through repeated bottom-up and top-down processing [21]. While Hourglass models are less common than U-Net in cervical cytology research, they offer an alternative localisation method that emphasises point-level predictions rather than segmentation. This justifies the inclusion of Stacked Hourglass as a localisation model alongside U-Net in this research.

D. Model Performance

Interpretability is important in medical artificial intelligence because users need to understand why a model produces a prediction [7], [8]. In this research, interpretability is supported through confusion matrices, per-class F1 comparisons, localisation heatmaps, extracted keypoints and integrated prediction examples. These visual outputs help show the strengths and weaknesses of each model more clearly than tables alone.

III. PROPOSED FRAMEWORK

A. Data Preparation

RIVA images and JSON keypoint annotation files were organised before model development. Each image was assigned an image-level label based on the labels found in its annotation file. For binary classification, an image was treated as pre-cancerous if at least one pre-cancerous label was present. Images without pre-cancerous labels were treated as non-lesion. Under this binary setting, 623 images were assigned as pre-cancerous and 336 images were assigned as non-lesion.

For multiclass classification, three label settings were prepared. The six-class setting included ASCUS, LSIL, ASCH, HSIL, SCC and INFL. The seven-class setting added ENDO. The eight-class setting included all available labels by adding NILM. Since some images contained more than one label, priority rules were used to assign one image-level label for multiclass training.

The dataset was divided into training, validation and testing subsets using a 70:15:15 split. Stratified splitting was applied to preserve class representation where possible. Images were resized to 224 x 224 pixels and normalised using ImageNet statistics. Data augmentation such as flipping, rotation, colour jittering and affine transformation was used during training.

Table 1 Attribute Table for Dataset

Attribute	Description	Type / Values / Format
Dataset Name	Name of the dataset	RIVA
Total Images	Number of mini patches	959 SVS files using Aperio Image Scope converted into PNG
Cell Labels	Pre-cancerous diagnostic categories	ASCH, ASCUS, HSIL, LSIL, SCC
	Non-lesion diagnostic categories	NILM, ENDO, INFL
File Names	Sample image file names	SCC_1_1.png, HSIL_1_1.png,

		ASCH_1_1.png, LSIL_1_1.png
Multi-Expert Subset	Images annotated by multiple experts (four cytologists)	386 images 17716 annotations
Single-Expert Subset	Images annotated by one cytologist	573 images 8442 annotations
Total Annotations	Number of cell annotations	26158
Unique Cells	Distinct cells in dataset	15949
Annotation Method	How annotations were made	Keypoint labelling using Label Studio
Scan Magnification	Microscope magnification	40x magnification using Grundium Ocus 40
Resolution	Resolution per image	0.25 μm /pixel resolution
Number of Pixels	Number of pixels per image	1024 x 1024
Source	Sample origin	115 patients from the Hospital Bernardino Rivadavia in Buenos Aires, Argentina
Data Split	Partitioning for model training/testing	Train : 70% Validation: 15% Test : 15%
Data Format	File types	Images : PNG Annotations: JSON

B. Classification Model Development

Three classification models were developed and compared. The baseline CNN used convolutional layers, ReLU activation, pooling, dropout and a fully connected classifier. ResNet34 and EfficientNet-B0 were implemented using transfer learning, where the final classification layer was modified according to the number of classes in the task.

For binary classification, weighted binary cross-entropy with logits was used to reduce the effect of class imbalance. For multi-class classification, weighted sampling and focal loss were used so that minority classes contributed more strongly during training. Hyperparameter tuning explored learning rate, weight decay, dropout and optimiser settings. Adam or AdamW optimisation, learning-rate scheduling and early stopping were used for training.

The classification models were evaluated using accuracy, precision, recall, F1-score, macro F1-score and confusion matrices. F1-score and macro F1-score were emphasised because accuracy alone would hide weak minority-class performance in imbalanced datasets.

C. Localisation Model Development

For localisation, keypoint annotations were converted into heatmap targets. U-Net and Stacked Hourglass were trained to predict heatmaps showing likely pre-cancerous cell locations.

Table 3 Multi-class Classification Performance Evaluation

Task	Model	Accuracy	Precision	Recall	F1	Macro F1
Six-class	CNN	0.0952	0.0159	0.1667	0.0290	0.0290
	ResNet34	0.2778	0.2683	0.2451	0.2503	0.2503
	EfficientNet-B0	0.3175	0.3063	0.2935	0.2926	0.2926
Seven-class	CNN	0.0233	0.0033	0.1429	0.0065	0.0065
	ResNet34	0.2946	0.4521	0.3344	0.3361	0.3361

U-Net was selected because its encoder-decoder structure is suitable for preserving spatial information. Stacked Hourglass was selected because its multi-scale refinement is suitable for keypoint-style prediction.

Localisation performance was evaluated using heatmap-level and point-level metrics. Heatmap-level metrics included test loss, mAP per image, global Average Precision, precision, recall, F1-score, Dice score and IoU. Point-level calibration was performed using different heatmap thresholds, kernel sizes and maximum point settings. This was done to convert heatmaps into extracted keypoints and compare them against ground-truth keypoint positions.

The best binary classifier was integrated with the localisation stage to improve interpretability. EfficientNet-B0 was selected as the binary classifier because it achieved the best binary classification result. The model was used to support the final point-level prediction pipeline by combining image-level abnormal prediction with spatial localisation output.

IV. RESULTS

A. Classification Results

The binary classification results in Table 2 and Fig. 1 show that EfficientNet-B0 achieved the best overall performance for classifying pre-cancerous and non-lesion images. CNN achieved high precision but low recall. This means that the model was less likely to produce false positives but missed a large number of actual pre-cancerous images which is not ideal in a screening context because false negatives may lead to missed abnormal cases.

ResNet34 has improved recall and F1-score showing that the residual architecture was able to capture more useful visual features. However, EfficientNet-B0 achieved the best balance between precision and recall, resulting in the highest F1-score and macro F1-score. This supports its selection as the best binary classification model in this research.

Table 2 Binary Classification Results

Model	Precision	Recall	F1	Macro F1
CNN	0.862	0.269	0.41	0.683
ResNet34	0.877	0.538	0.67	0.816
EfficientNet-B0	0.766	0.774	0.77	0.866

Table 3 and Fig. 1 shows that for multi-class classification EfficientNet-B0 achieved the best performance in the six-class setting while ResNet34 performed better for the seven-class and eight-class settings. This demonstrates that no model performed best across all multi-class settings.

The overall multi-class classification performance was lower than binary classification performance. This is expected because multi-class classification requires the models to distinguish between multiple visually similar cervical cell categories. Classes such as ASCUS, ASCH, LSIL and HSIL may share similar morphological features. In addition, minority classes such as ASCUS, ASCH, SCC and ENDO contain fewer samples compared with majority classes such as NILM and INFL.

	EfficientNet-B0	0.2791	0.2746	0.2587	0.2351	0.2351
Eight-class	CNN	0.0903	0.0840	0.1458	0.0494	0.0494
	ResNet34	0.2639	0.3719	0.3143	0.3047	0.3047
	EfficientNet-B0	0.2917	0.2544	0.2669	0.2482	0.2482

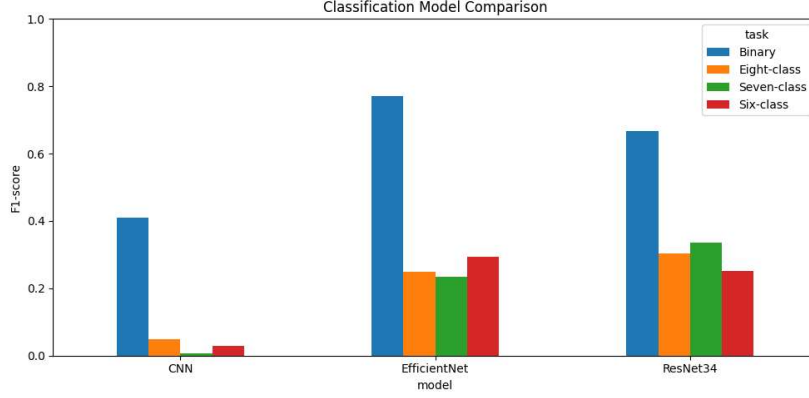


Figure 1 F1 score of Binary and Multi-class Classification Model Comparison

B. Localisation Results

The localisation results in table 4 and Fig. 2 show that both U-Net and Stacked Hourglass were able to generate heatmap-based localisation outputs and extracted keypoints for pre-cancerous cervical cells. U-Net achieved slightly lower test loss, which indicates stable heatmap regression during evaluation. However, Stacked Hourglass achieved slightly stronger overall localisation performance.

The difference between U-Net and Stacked Hourglass was not extremely large, which shows that both models are usable for localisation. However, Stacked Hourglass performed

slightly better overall. This suggests that its multi-scale refinement process is more suitable for keypoint-style cell localisation from heatmap outputs.

The final integrated prediction examples in Fig. 3 compare the original image, ground-truth keypoints and the integrated model heatmap output. This is important because localisation models should not only achieve numerical performance.

Fig. 3 shows that both integrated U-Net and Stacked Hourglass pipeline can highlight suspected regions in cervical cytology images. Some activations are focused around relevant abnormal-cell regions, while other activations are broader around stained or visually complex areas. This shows that the model is useful for visual interpretation.

Table 4 Localisation Performance Evaluation

Model	Test Loss	mAP per Image	Global AP	Precision	Recall	F1	Dice	IoU
U-Net	0.5874	0.5101	0.3615	0.3049	0.5625	0.3955	0.3955	0.2465
Stacked Hourglass	0.5979	0.5104	0.3725	0.3193	0.5629	0.4075	0.4075	0.2559

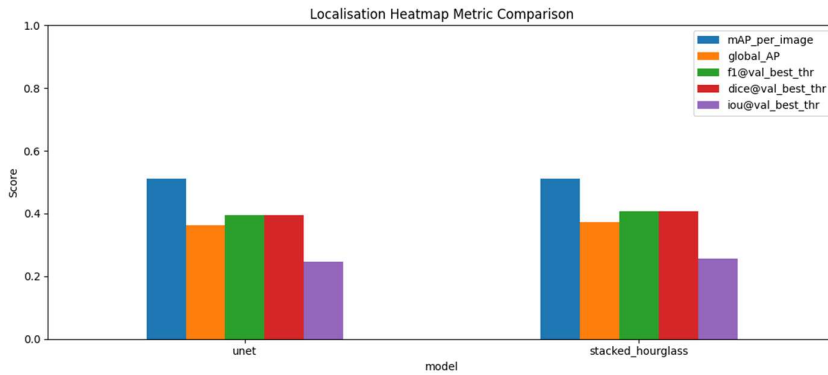


Figure 2 Localisation Heatmap Metric Comparison

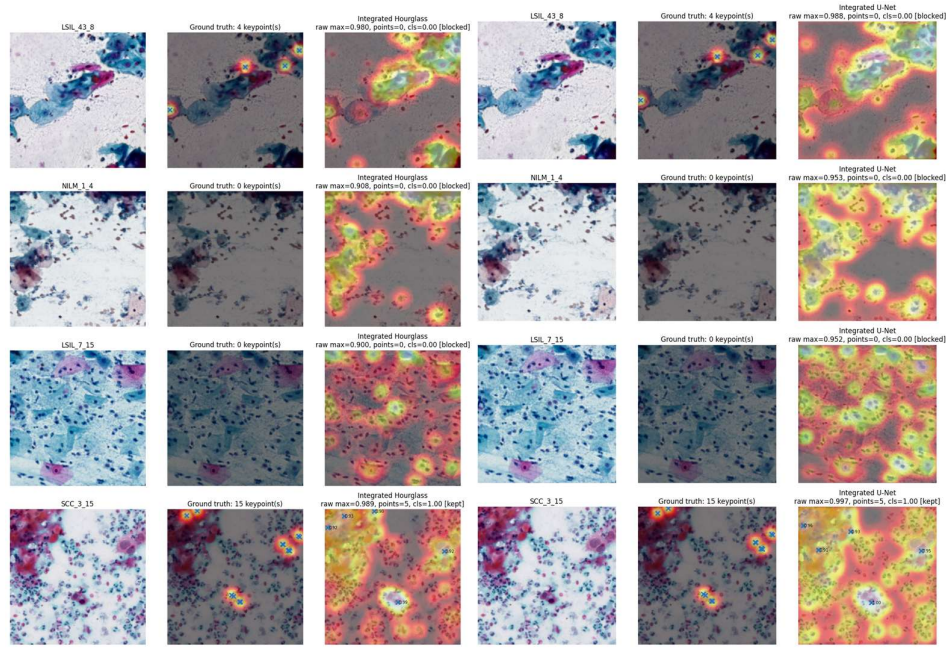


Figure 3 Final Integrated U-Net and Stacked Hourglass Point-Gate Prediction Examples

V. DISCUSSION

The binary classification results show that EfficientNet-B0 is the most suitable classifier for the image-level screening task in this project. Its balanced precision and recall are important because false negatives are especially undesirable in medical screening. CNN was too conservative and missed many abnormal images, while ResNet34 improved recall but did not achieve the same overall balance as EfficientNet-B0.

The stronger performance of EfficientNet-B0 can be explained by its compound scaling. This allows the model to balance network depth, width and image resolution, making it more suitable for a limited and imbalanced medical imaging dataset. Therefore, EfficientNet-B0 was selected as the best binary classification model and was used in the final integrated pipeline.

The multiclass results show that classification becomes more difficult when the number of diagnostic labels increases. EfficientNet-B0 performed best for the six-class setting, while ResNet34 performed best for the seven-class and eight-class settings. This shows that model performance can change depending on the number of classes and the type of labels included. The confusion matrices further show that CNN was not suitable for multiclass classification because its predictions were concentrated in limited classes.

For localisation, U-Net and Stacked Hourglass both produced usable heatmap outputs. U-Net achieved lower test loss, but Stacked Hourglass performed better across most heatmap-based and point-level metrics. This indicates that lower loss alone does not guarantee better localisation quality. Since Stacked Hourglass achieved slightly higher heatmap-based metrics, it was selected as the stronger overall localisation model.

Overall, the results indicate that EfficientNet-B0 is suitable for binary cervical cytology classification, ResNet34 is stronger for more complex multiclass classification, and

Stacked Hourglass is slightly better for pre-cancerous cell localisation. The findings also show that class imbalance, multi-label images and visual similarity between diagnostic categories remain major challenges in this research.

VI. LIMITATIONS AND FUTURE WORKS

There are several limitations in this research. First, the RIVA dataset is imbalanced, meaning that some classes contain far fewer samples than others. This affected the model performance especially for minority classes such as ASCUS, ASCH, SCC and ENDO. Second, some images contain multiple labels, but the multiclass classification task required one image-level label to be assigned. This may simplify the real condition of cervical cytology images.

Third, localisation was performed using keypoint annotations rather than full cell masks. Therefore, the models learned cell-centre heatmaps instead of complete cell boundaries. Although this supports point-level localisation, it does not provide full segmentation of abnormal cells. The final integrated model also remains as a research prototype and should not be used for clinical diagnosis without further validation.

Future work should improve minority-class detection through targeted augmentation, synthetic sample generation or more advanced sampling strategies. Future studies can also investigate more advanced architectures such as Vision Transformers, hybrid CNN-transformer models, object detection models or segmentation models. External validation on additional cervical cytology datasets is also recommended to evaluate generalisation ability.

VII. CONCLUSION

This research developed and evaluated a deep learning-based framework for cervical cytology image classification and localisation using the RIVA dataset. The study focused on comparing multiple classification architectures, extending the classification task into multiclass settings, developing

localisation models, and interpreting model outputs using visualisation and analytical techniques.

For Objective 1, binary classification was performed to distinguish pre-cancerous and non-lesion cervical cytology images. The results showed that EfficientNet-B0 achieved the best overall binary classification performance compared to CNN and ResNet34. For Objective 2, multiclass classification was carried out using six, seven and eight label settings. The results showed that EfficientNet-B0 achieved the best performance for the six-class setting, while ResNet34 achieved the best performance for the seven-class and eight-class settings based on F1-score and macro F1-score.

For Objective 3, localisation models were developed using U-Net and Stacked Hourglass. Both models generated heatmap-based localisation outputs and extracted keypoints for pre-cancerous cell localisation. Stacked Hourglass achieved slightly better overall localisation performance compared to U-Net based on heatmap-based metrics and point-level calibration results. The best binary classification model was also integrated as a point-level classifier gate during the final integrated prediction stage, allowing the system to combine image-level classification with localisation outputs.

For Objective 4, visualisation and analytical techniques were used to interpret the results. Confusion matrices were used to analyse misclassification patterns, while localisation heatmaps, extracted keypoints and prediction examples were used to examine whether the localisation outputs corresponded to abnormal cell regions. In conclusion, this research achieved its objectives by developing and comparing deep learning models for cervical cytology image classification and localisation.

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